

MATH 123 LECTURE NOTES

WEEK VII

EXTREME VALUES

Absolute Extreme Values

→ Function f has absolute maximum value $f(x_0)$ at point x_0 in its domain if $f(x) \leq f(x_0)$ holds for every x in the domain of f .

→ Similarly f has absolute minimum value $f(x_1)$ at point x_1 in its domain if $f(x) \geq f(x_1)$ holds for every x in the domain of f .

THEOREM (Existence of extreme values)

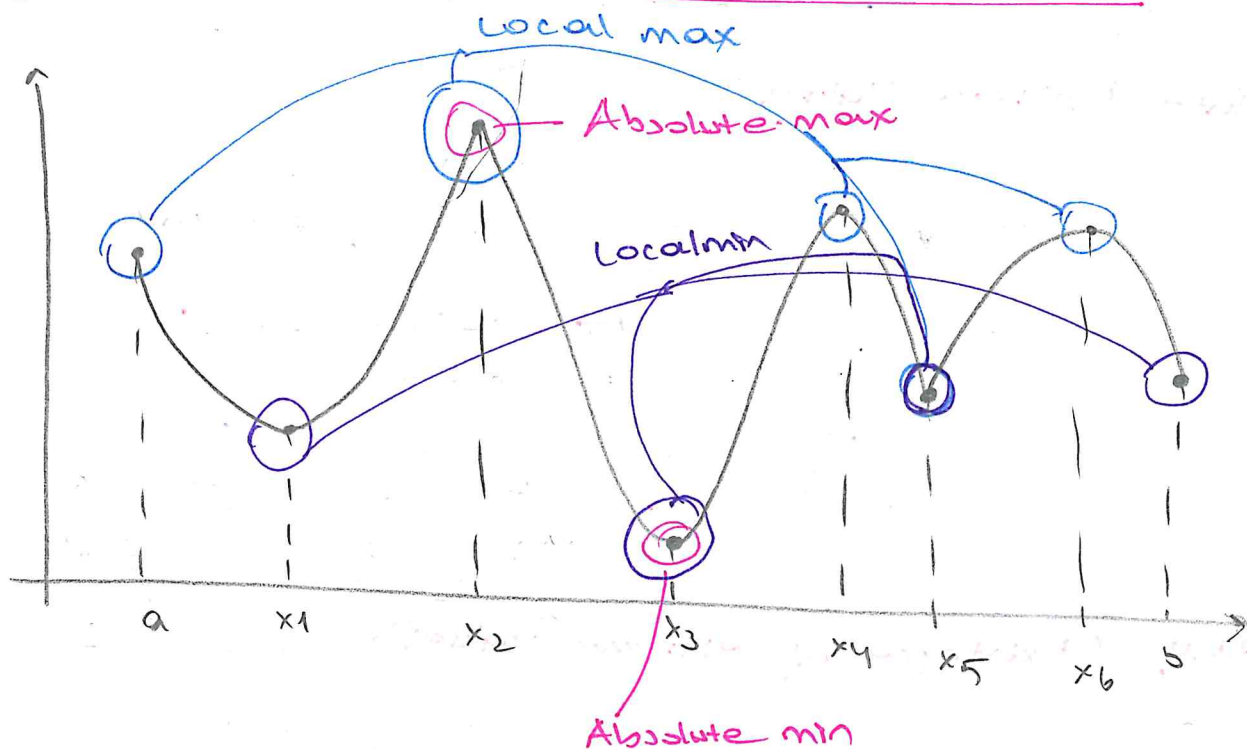
If the domain of the function f is closed, finite interval or a union of finitely many such intervals and if f is continuous on that domain, then f must have an absolute maximum value and an absolute minimum value.

Local Extreme Values

→ Function f has a local maximum value (local max) $f(x_0)$ at the point x_0 in its domain provided that there exists a number $h > 0$ such that $f(x) \leq f(x_0)$ whenever x is in the domain of f and $|x - x_0| < h$.

→ Similarly, f has a local minimum value (loc. min) $f(x_1)$

at the point x_1 in its domain provided there exists a number $h > 0$ such that $f(x) \geq f(x_1)$ whenever x is in the domain of f and $|x - x_1| < h$



f has local maximum values at a, x_2, x_4, x_6

f has local minimum values at x_1, x_3, x_5, b

Critical Points, Singular Points, and End Points

$f(x)$ can have local extreme values only at points x of three special types:

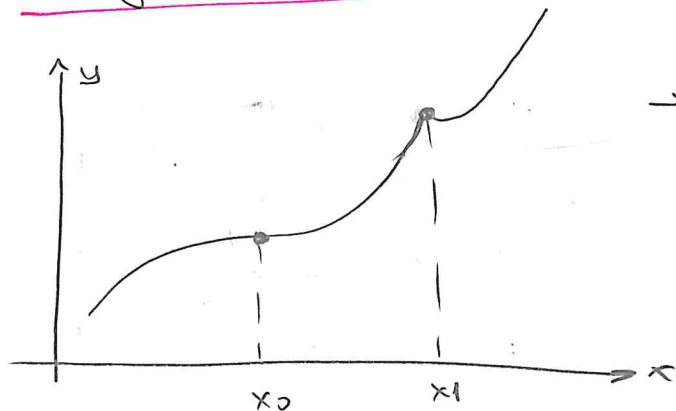
① critical points points in $x \in D(f)$ where $f'(x) = 0$

② Singular points points in $x \in D(f)$ where $f'(x)$ NOT defined.

③ endpoints points in $D(f)$ that do not belong to any open interval contained in $D(f)$

Thm Locating extreme values

If the function f is defined on an interval I and has a local maximum (or local minimum) value at point $x=x_0$ in I , then x_0 must be either a critical point of f , a singular point of f , or an endpoint of I .



→ Although a function can have extreme values anywhere other than at endpoints, critical points and singular points. This figure shows that a function with critical point x_0 and a singular point x_1 at neither of which has an extreme value.

Ex 7 Find the maximum and minimum values of the function $g(x) = x^3 - 3x^2 - 9x + 2$ on the interval $-2 \leq x \leq 2$

Since $g(x)$ is a polynomial, it doesn't have singular points ($x \in D(g)$ $g'(x)$ not defined.)

$$g'(x) = 3x^2 - 6x - 9 = 3(x^2 - 2x - 3) = 3(x+1)(x-3) = 0$$

If $x = -1$ or $x = 3 \rightarrow$ not in the domain.

So we have

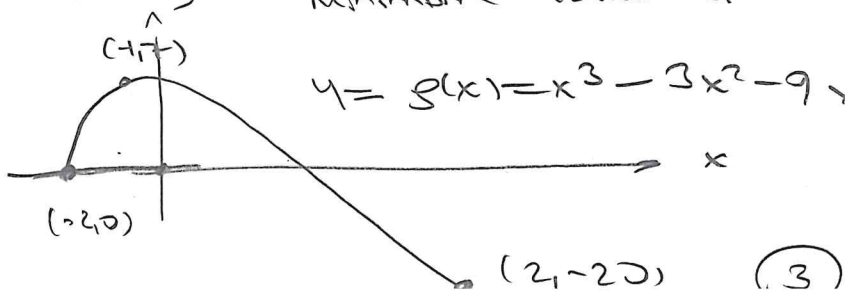
Critical point $x = -1$ $g(-1) = 7$

End points $\begin{cases} x = -2 & g(-2) = 0 \\ x = 2 & g(2) = -20 \end{cases}$

Absolute Maximum value at $x = -1$ $g(-1) = 7$.

Absolute Minimum value at $x = 2$ $g(2) = -20$

$$y = g(x) = x^3 - 3x^2 - 9x + 2$$



Ex 7 Find minimum and maximum values of $h(x) = 3x^{2/3} - 2x$ on the interval $[-1, 1]$

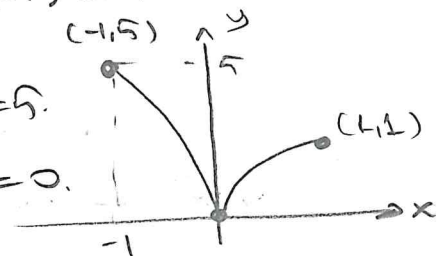
$$h'(x) = 3 \cdot \frac{2}{3} x^{-1/3} - 2 = 2(x^{-1/3} - 1)$$

Singular point $x=0$ $h'(x)$ is not defined $\Rightarrow h(0)=0$

Critical point $x=1$ $h'(1)=0 \Rightarrow h(1)=1$

End points $x=-1$ $x=1 \Rightarrow h(-1)=5$ $h(1)=1$

Absolute Maximum value at $x=-1$ $h(-1)=5$
 Absolute Minimum value at $x=0$ $h(0)=0$



THM The First Derivative Test.

PART I Testing interior critical points and singular points

Suppose that f is continuous at x_0 , and x_0 is not an endpoint of the domain of f .

(a) If there exists an open interval (a, b) containing x_0 such that $f'(x) > 0$ on (a, x_0) and $f'(x) < 0$ on (x_0, b) then

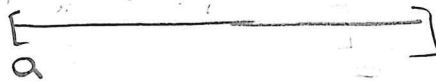
f has a local maximum value at x_0 .

(b) If there exists an open interval (a, b) containing x_0 such that $f'(x) < 0$ on (a, x_0) and $f'(x) > 0$ on (x_0, b) then

f has a local minimum value at x_0 .

PART II Testing end points of the domain.

Suppose a is a left endpoint of the domain of f and f is right continuous at a .



(c) If $f'(x) > 0$ on some interval (a, b) , then f has local minimum at a .

(d) If $f'(x) < 0$ on some interval (a, b) , then f has local maximum at a .

Suppose b is a right endpoint of the domain of f and f is left continuous at b .



(e) If $f'(x) > 0$ on some interval (a, b) , then f has a local maximum at b .

(f) If $f'(x) < 0$ on some interval (a, b) , then f has a local minimum at b .

Ex Find the local and absolute extreme values of $f(x) = x^4 - 2x^2 - 3$ on the interval $[-2, 2]$. Sketch the graph of f .

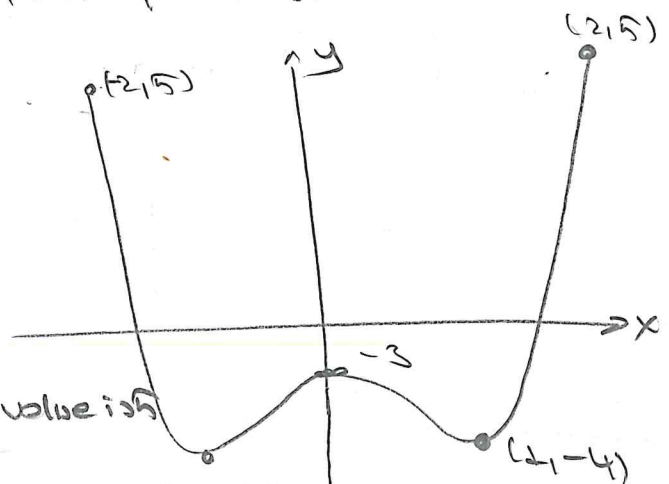
$$f'(x) = 4x^3 - 4x = 4(x^3 - x) = 4x(x-1)(x+1)$$

Critical points $x=0, x=1, x=-1$ $f(0) = -3$ $f(1) = f(-1) = -4$

Singular points No singular points.

End points $x=-2, x=2$ $f(2) = f(-2) = 5$.

x	-2	-1	0	1	2
f'	-	+	-	+	
f	max	min	max	min	max



Absolute max at points $x=-2$ value is 5
 $x=2$

Absolute min at point $x=-1$ value is -4
 $x=1$

Ex Find and classify the local and absolute extreme values of the function $f(x) = x - x^{2/3}$ with domain $[-1, 2]$. Sketch the graph of f

$$f'(x) = 1 - \frac{2}{3} x^{-1/3} = \left(x^{1/3} - \frac{2}{3} \right) / x^{1/3}$$

Singular point $x=0 \Rightarrow f(0)=0$

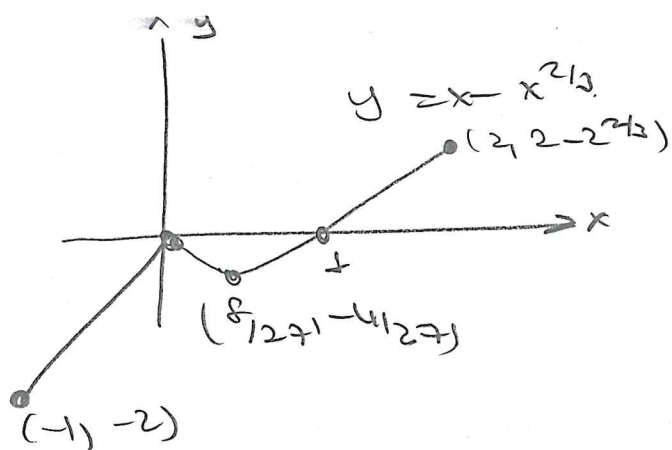
Critical point $f'(x)=0 \quad x = 8/27 \Rightarrow f(8/27) = -4/27$

Endpoints $x=-1 \quad f(-1) = -2 \quad (\text{ABS MIN})$

$x=2 \quad f(2) = 2 - 2^{2/3} \approx 0.4126 \quad (\text{ABS MAX})$

x	-1		0		$8/27$		2
f'		+	undef	-	0	+	
f		min		max		min	max
		ABS MIN (loc)		local max		local min	ABS MAX (loc)

(If it's abs, mention it's local)



FUNCTIONS NOT DEFINED ON CLOSED, FINITE INTERVAL

THM Existence of extreme values on open intervals

If f is continuous on the open interval (a, b) , and if

$$\lim_{x \rightarrow a^+} f(x) = L \quad \text{and} \quad \lim_{x \rightarrow b^-} f(x) = M$$

then the following conclusions hold.

(i) If $f(w) > L$ and $f(v) > M$ for some $w, v \in (a, b)$, then f has an absolute maximum value on (a, b)

(ii) If $f(v) < L$ and $f(v) < M$ for some $v \in (a, b)$, then f has an absolute minimum value on (a, b)

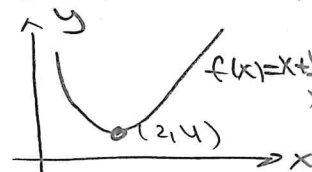
NOTE If $a = -\infty$ we check $\lim_{x \rightarrow -\infty}$ instead $\lim_{x \rightarrow a^+}$

$b = \infty$ we check $\lim_{x \rightarrow \infty}$

Also, either or both L and M may be either ∞ or $-\infty$

Ex Show that $f(x) = x + 4/x$ has an absolute minimum value on the interval $(0, \infty)$ and find that minimum

$$\lim_{x \rightarrow 0^+} f(x) = \infty \quad \text{and} \quad \lim_{x \rightarrow \infty} f(x) = \infty$$



Since $f(2) = 5 < \infty$, Theorem guarantees that

f must have an absolute minimum value at some point in $(0, \infty)$

x	0	2	
f'	-	0	+

$$f'(x) = 1 - \frac{4}{x^2} = \frac{x^2 - 4}{x^2} = \frac{(x-2)(x+2)}{x^2}$$

not indeterminate

which equals 0 only at $x=2$ and $x=-2$.

Since f has $(0, \infty)$ domain. It has no singular points, only one critical value $x=2$ where $f(2)=4$.

This must be absolute min value of f .

Ex Let $f(x) = x \cdot e^{-x^2}$. Find and classify the critical points of f , evaluate $\lim_{x \rightarrow \pm\infty} f(x)$

$$f'(x) = e^{-x^2} (1 - 2x^2) = 0 \Rightarrow 1 - 2x^2 = 0 \Leftrightarrow$$

The critical points are $x = \pm \frac{1}{\sqrt{2}}$ $f(\pm \frac{1}{\sqrt{2}}) = \pm \frac{1}{\sqrt{2}e}$

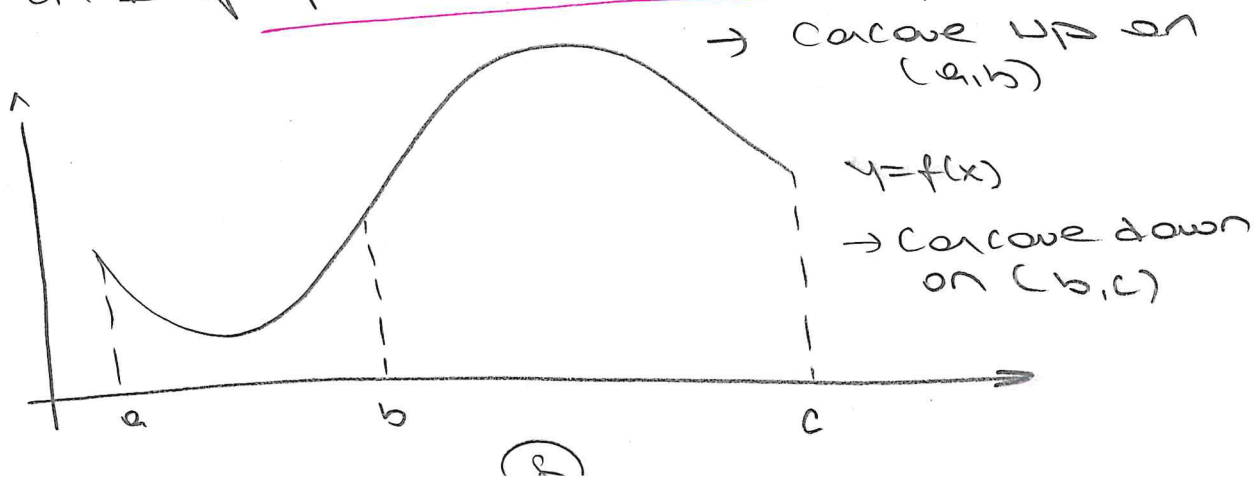
x		$-\frac{1}{\sqrt{2}}$		$\frac{1}{\sqrt{2}}$	
f'	-	0	+	0	-
		min		max	

$$\lim_{x \rightarrow \pm\infty} x \cdot e^{-x^2} = \lim_{x \rightarrow \pm\infty} \frac{x}{e^{x^2}} \stackrel{\text{L'Hopital}}{=} \lim_{x \rightarrow \pm\infty} \frac{1}{2xe^{x^2}} = 0$$

$f(\frac{1}{\sqrt{2}}) > 0$ and $f(-\frac{1}{\sqrt{2}}) < 0 \Rightarrow f$ assumes both absolute max and absolute min value.

CONCAVITY AND INFLECTIONS

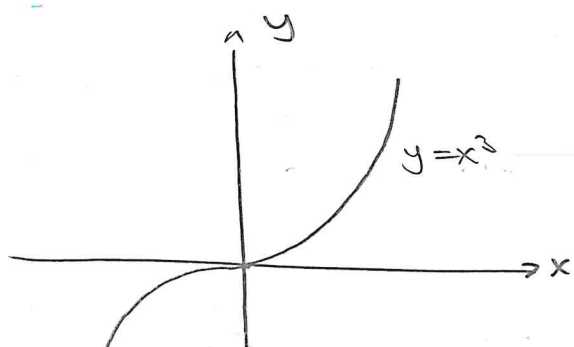
$\rightarrow$ We say that the function f is concave up on an open interval I if it is differentiable there and the derivative f' is an increasing function on I . Similarly, f is concave down on I if f' exists and is decreasing on I .



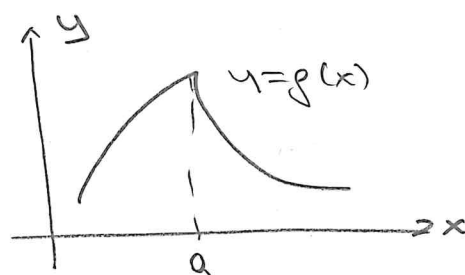
Inflection Points

We say that the point $(x_0, f(x_0))$ is an **inflection point** of the curve $y=f(x)$ (or that the function f has an inflection point at x_0) if the following two conditions are satisfied:

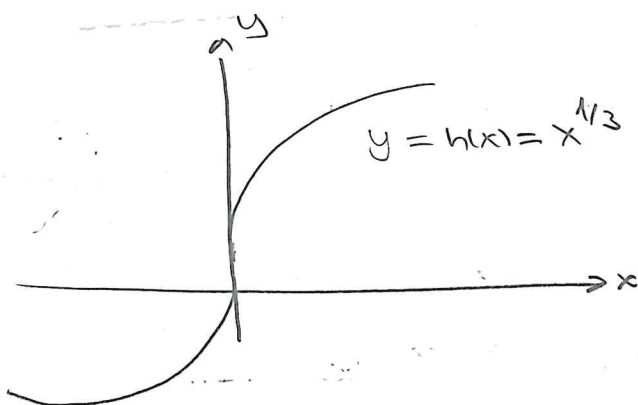
- (a) the graph of $y=f(x)$ has a tangent line at $x=x_0$
- (b) the concavity of f is opposite on opposite sides of x_0



→ $x=0$ is a critical point of $y=x^3$ and f has an inflection point at that point



→ The concavity of g is opposite on opposite sides of point a , but there's no tangent at a therefore a is not an inflection point



→ The graph of h has an inflection point at the origin even though $x=0$ is a singular point of h

THM CONCAVITY AND THE SECOND DERIVATIVE

- (a) If $f''(x) > 0$ on interval I , then f is concave up on I
- (b) If $f''(x) < 0$ on interval I , then f is concave down on I
- (c) If f has an inflection point at x_0 and $f''(x_0)$ exists, then $f''(x_0) = 0$

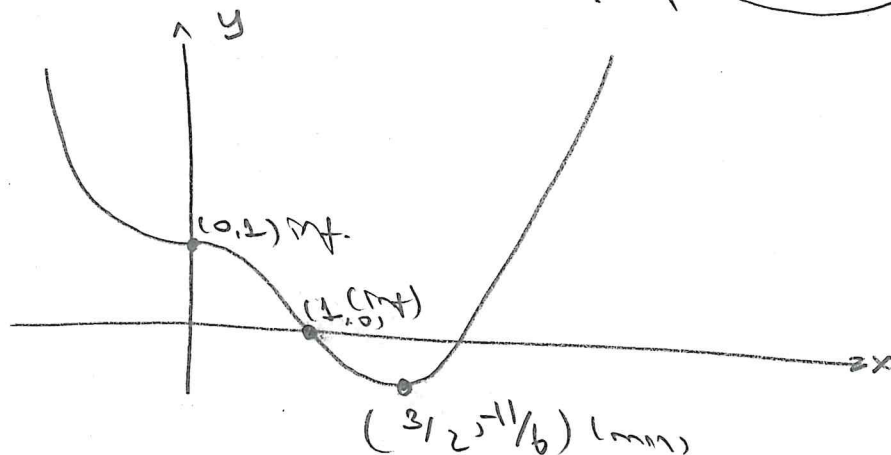
Ex 2 Determine the intervals of increase and decrease, the local extreme values, and the concavity of $f(x)$

$f(x) = x^4 - 2x^3 + 1$. Sketch the graph.

$$f'(x) = 4x^3 - 6x^2 = 2x^2(2x - 3) = 0 \quad x = 0 \quad x = 3/2$$

$$f''(x) = 12x^2 - 12x = 12x(x - 1) \quad x = 0 \quad x = 1$$

x	0	1	$3/2$
f'	—	—	+
f''	+	—	+
f			



THE SECOND DERIVATIVE TEST

- (a) If $f'(x_0) = 0$ and $f''(x_0) < 0$, then f has a local max at x_0
- (b) If $f'(x_0) = 0$ and $f''(x_0) > 0$, then f has local min at x_0
- (c) If $f'(x_0) = 0$ and $f''(x_0) = 0$, no conclusion can be drawn; f may have a local maximum at x_0 or a local minimum, or it may have an inflection point instead.

Ex Find and classify the first derivatives of f :

$$f'(x) = (2x - x^2)e^{-x} = x(2-x)e^{-x} = 0 \quad x=0 \text{ and } x=2$$

$$f''(x) = (2 - 4x + x^2)e^{-x}$$

$$f''(0) = 2 > 0 \rightarrow \text{local min } x=0$$

$$f''(2) = -2e^{-2} < 0 \text{ local max at } x=2$$

ASYMPTOTES

Definition The graph of $y = f(x)$ has a vertical asymptote at $x=a$ if either

$$\lim_{x \rightarrow a^-} f(x) = \pm \infty \text{ or } \lim_{x \rightarrow a^+} f(x) = \pm \infty \text{ or both.}$$

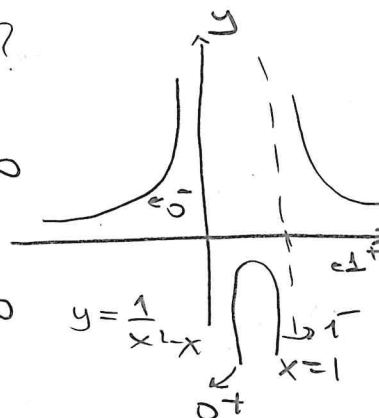
Ex Find the vertical asymptotes of $f(x) = \frac{1}{x^2 - x}$. How does the graph approach these asymptotes?

$$\lim_{x \rightarrow 0^-} \frac{1}{x^2 - x} = \infty$$

$$\lim_{x \rightarrow 1^-} \frac{1}{x(x-1)} = -\infty$$

$$\lim_{x \rightarrow 0^+} \frac{1}{x^2 - x} = -\infty$$

$$\lim_{x \rightarrow 1^+} \frac{1}{x(x-1)} = \infty$$



Def The graph $y=f(x)$ has a horizontal asymptote at $y=b$ if either

$$\lim_{x \rightarrow \infty} f(x) = b \quad \text{or} \quad \lim_{x \rightarrow -\infty} f(x) = b$$

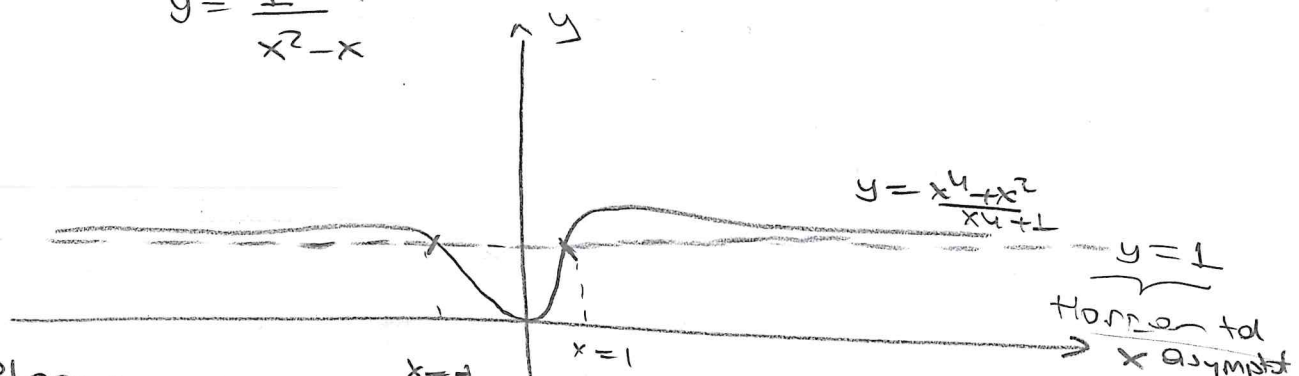
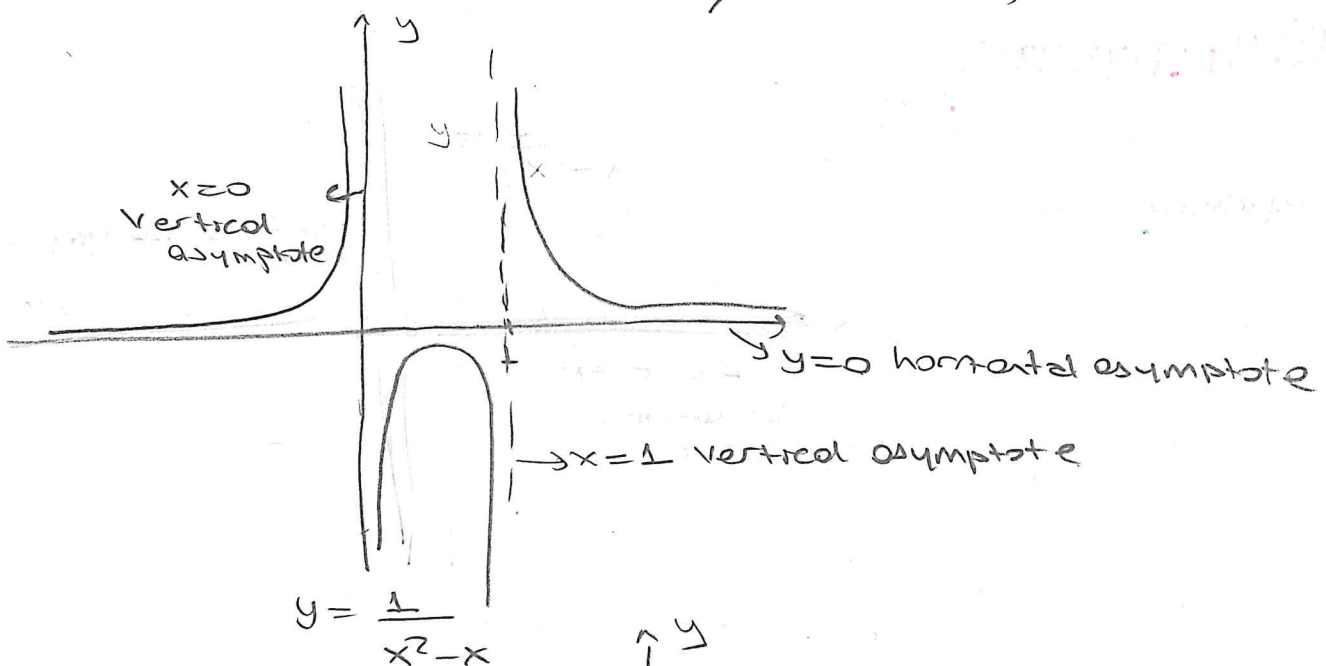
Ex 7 Find the horizontal asymptotes of

① $f(x) = \frac{1}{x^2 - x}$

② $p(x) = \frac{x^4 + x^2}{x^4 + 1}$

① $\lim_{x \rightarrow \pm\infty} \frac{1}{x^2(1 - \frac{1}{x})} = \lim_{x \rightarrow \pm\infty} \frac{1/x^2}{1 - \frac{1}{x}} = \frac{0}{1} = 0$ $y=0$ horizontal asymptote

② $\lim_{x \rightarrow \pm\infty} \frac{x^4 + x^2}{x^4 + 1} = \lim_{x \rightarrow \pm\infty} \frac{x^4(1 + 1/x^2)}{x^4(1 + 1/x^4)} = 1$ $y=1$ horizontal asymptote



NOTE Please note that graph of $p(x)$ crosses its asymptote twice.

Ex 7 Find vertical and horizontal asymptotes of $f(x) = \frac{\sin x}{x}$

$x=0$ is a candidate to be a vertical asymptote

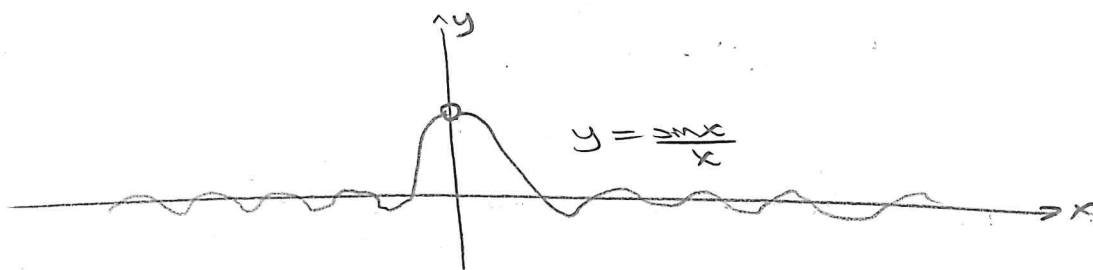
$$\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1 \neq \pm \infty \quad \text{so } x=0 \text{ is NOT a vertical asymptote.}$$

$$\lim_{x \rightarrow \infty} \frac{\sin x}{x} = 0, \quad \lim_{x \rightarrow -\infty} \frac{\sin x}{x} = 0 \quad \text{Sandwich theorem} \quad -1 \leq \sin x \leq 1$$

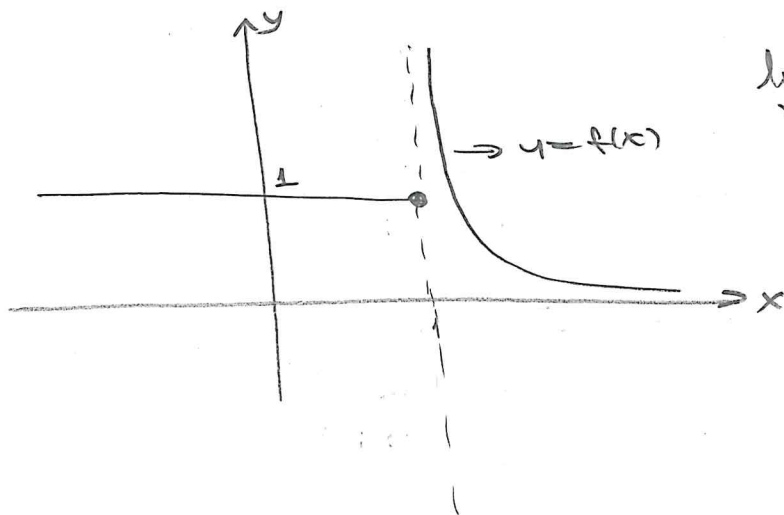
$$y=0 \text{ is a horizontal asymptote. } \left. \begin{array}{l} \lim_{x \rightarrow \pm \infty} -\frac{1}{x} = 0 \\ \lim_{x \rightarrow \pm \infty} \frac{1}{x} = 0 \end{array} \right\} \text{Sandwich Theorem } \lim_{x \rightarrow \pm \infty} \frac{\sin x}{x} = 0$$

$$-\frac{1}{x} \leq \frac{\sin x}{x} \leq \frac{1}{x}$$

REMARK This example shows that the x values which makes $f(x)$ undefined are NOT ALWAYS VERTICAL ASYMPTOTES.



Ex 7



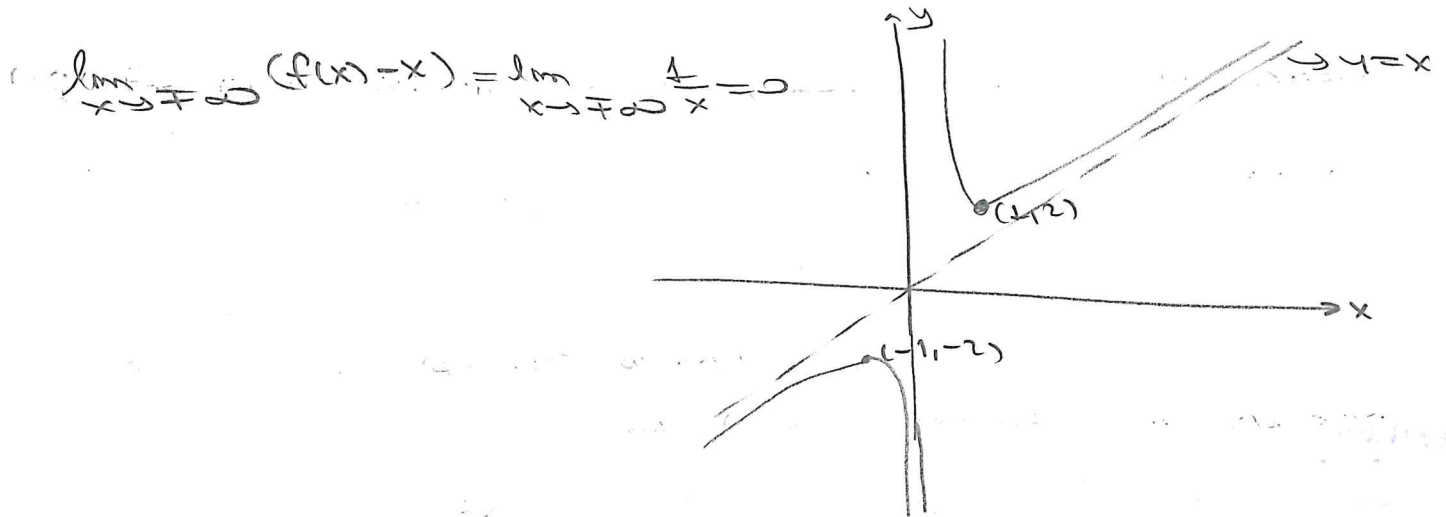
$\lim_{x \rightarrow 1^+} f(x) = \infty$ $x=1$ is a vertical asymptote from right.
Although $\lim_{x \rightarrow 1^-} f(x) = 1 \neq \pm \infty$.

Def The straight line $y = ax + b$ (where $a \neq 0$) is an oblique asymptote of the graph of $y = f(x)$ if

either $\lim_{x \rightarrow -\infty} (f(x) - (ax + b)) = 0$ or

$\lim_{x \rightarrow \infty} (f(x) - (ax + b)) = 0$ or both

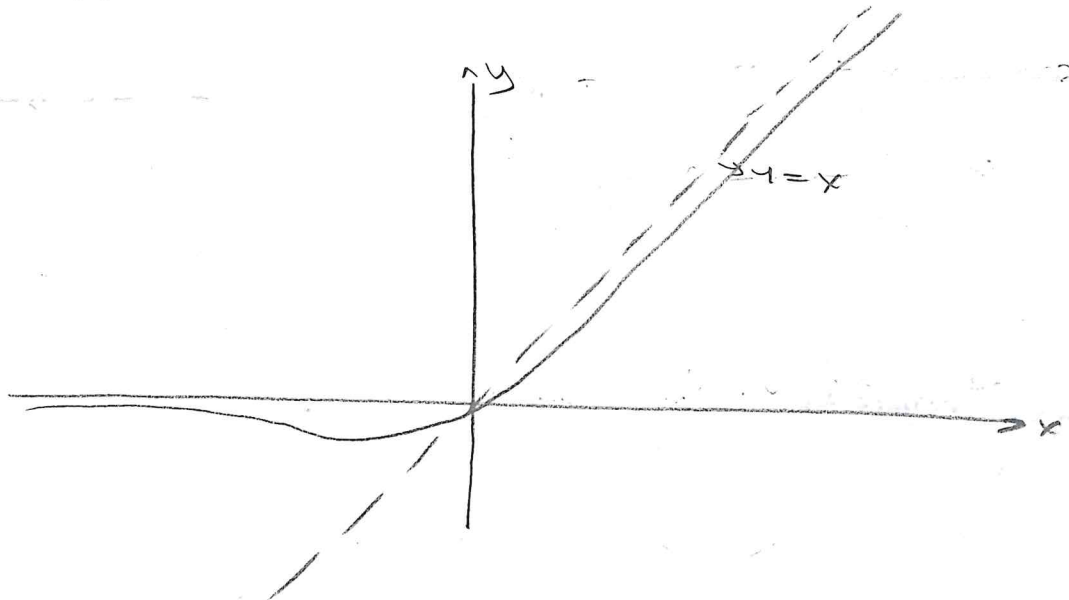
Ex 7 Consider the function $f(x) = \frac{x^2+1}{x} = x + \frac{1}{x}$ whose graph is shown. The straight line $y=x$ is a two-sided oblique asymptote of the graph of f because



Ex 7 The graph of $y = \frac{x \cdot e^x}{1+e^x}$ has a horizontal asymptote $y=0$ at the left and an oblique asymptote $y=x$ at the right.

$$\lim_{x \rightarrow -\infty} \frac{x \cdot e^x}{1+e^x} = \frac{0}{1} = 0 \quad \text{and}$$

$$\lim_{x \rightarrow \infty} \left(\frac{x \cdot e^x}{1+e^x} - x \right) = \lim_{x \rightarrow \infty} \frac{x(e^x - 1 - e^x)}{1+e^x} = \lim_{x \rightarrow \infty} \frac{-x}{1+e^x} = 0$$



ASYMPTOTES

Asymptotes of a rational function

Suppose that $f(x) = \frac{P_m(x)}{Q_n(x)}$, where P_m and Q_n are polynomials of degree m and n , respectively. Suppose also that P_m and Q_n have no common linear factors. Then

- (a) The graph of f has a vertical asymptote at every position x such that $Q_n(x) = 0$.
- (b) The graph of f has a two-sided horizontal asymptote $y = 0$ if $m < n$.
- (c) The graph of f has two-sided horizontal asymptote $y = L$ ($L \neq 0$) if $m = n$. L is the quotient of the coefficients of the highest degree terms in P_m and Q_n .
- (d) The graph of f has a two-sided oblique asymptote if $m = n + 1$. This asymptote can be found by dividing Q_n into P_m to obtain a linear quotient, $ax + b$, and remainder R , a polynomial of degree at most $n - 1$.
That's $f(x) = ax + b + \frac{R(x)}{Q_n(x)}$ The oblique asymptote is $y = ax + b$.
- (e) The graph of f has no horizontal or oblique asymptote if $m > n + 1$.

Ex Find the oblique asymptote of $y = \frac{x^3}{x^2 + x + 1}$

$$\begin{array}{r} x^3 \quad | \quad x^2 + x + 1 \\ x^3 + x^2 + x \\ \hline -x^2 - x - 1 \\ -x^2 - x - 1 \\ \hline 1 \end{array}$$

$$\frac{x^3}{x^2 + x + 1} = x - 1 + \frac{1}{x^2 + x + 1}$$

$y = x - 1$ is the oblique asymptote
(15)

CHECKLIST FOR CURVE SKETCHING

- ① Calculate $f'(x)$ and $f''(x)$ and express the results in factored form
- ② For $f(x)$
 - (a) Vertical asymptotes (look for denominators)
 - (b) Horizontal or oblique asymptotes ($\lim_{x \rightarrow \pm\infty} f(x)$)
 - (c) Roots $(x, 0)$ and $(0, y)$
 - (d) Symmetry
- ③ For $f'(x)$
 - (a) Critical points
 - (b) Any points f' is not defined (these points singular, endpoints, vertical asymptotes)
 - (c) Intervals on which f' is positive or negative
- ④ For $f''(x)$
 - (a) Points $f''(x) = 0$
 - (b) Points $f''(x)$ is undefined
 - (c) Intervals where f'' positive and negative
 - (d) Inflection points

Ex Sketch the graph of $y = \frac{x^2 + 2x + 4}{2x}$

Domain $\mathbb{R} \setminus \{0\}$

$f(x)$ Asymptotes

Vertical asymptote $x = 0$

$$\lim_{x \rightarrow 0^+} \frac{x^2 + 2x + 4}{x} = \infty$$

$$\lim_{x \rightarrow 0^-} \frac{x^2 + 2x + 4}{x} = -\infty$$

Oblique asymptote

$$y = \frac{x^2 + 2x + 4}{2x} = \frac{x}{2} + 1 + \frac{2}{x}$$

$$y - \left(\frac{x}{2} + 1\right) = \frac{2}{x}$$

$$\lim_{x \rightarrow \pm\infty} y - \left(\frac{x}{2} + 1\right) = \lim_{x \rightarrow \pm\infty} \frac{2}{x} = 0$$

Oblique asymptote

$$\frac{x}{2} + 1$$

Roots

$$y = \frac{x^2 + 2x + 4}{2x}$$

$$x^2 + 2x + 4 = 0$$

$$x_{1,2} = \frac{-2 \pm \sqrt{4 - 4 \cdot 4}}{2}$$

No real roots.

Symmetry

$$y(-x) = \frac{x^2 - 2x + 4}{-2x}$$

No relation with $y(x)$ No symmetry

$$f'(x) \quad y' = \frac{1}{2} - \frac{2}{x^2} = \frac{x^2 - 4}{2x^2}$$

$x=0$ y' undef asymptote
roots $x = \pm 2$

$$f(2) = 3$$

$$f(-2) = -1$$

$$f''(x) \quad y'' = \frac{4}{x^3}$$

$x=0$ y'' undef.

		CP.		ASY		CP.	
		-2		0		2	
x							
y'	+	0	-	undef	-	0	+
y''	-	-		undef	+	+	
y							

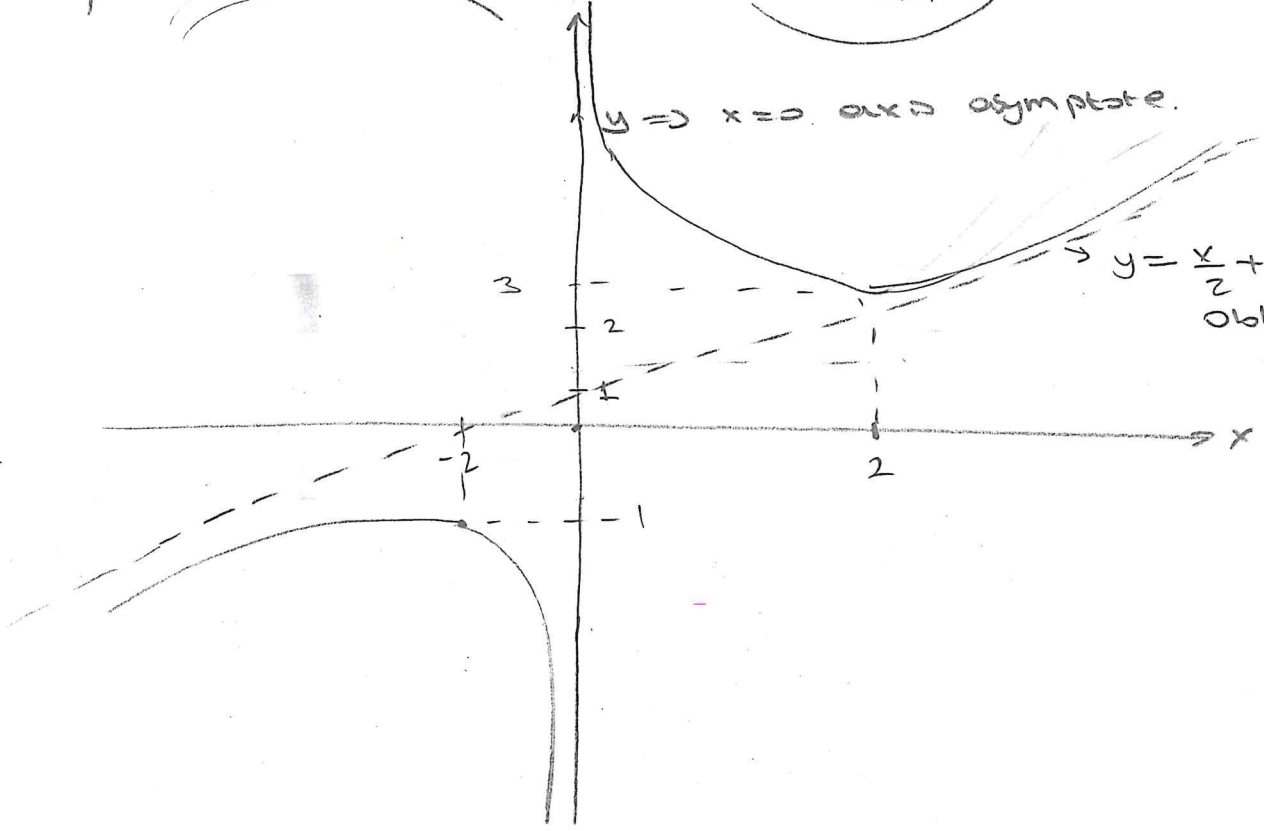
abs max

abs min

$y \Rightarrow x=0$ axis asymptote.

$$y = \frac{x}{2} + 1$$

Oblique asymptote



Ex Sketch the graph of $y = \frac{x^2-1}{x^2-4}$

$f(x)$ Domain $f = \mathbb{R} \setminus \{2, -2\}$

Asymptotes

$x=2$

$$\lim_{x \rightarrow 2^+} \frac{x^2-1}{x^2-4} = \infty$$

$$\lim_{x \rightarrow 2^-} \frac{x^2-1}{x^2-4} = -\infty$$

$$\frac{+}{2^-} \frac{+}{2^+}$$

Vertical asymptotes

$x=-2$

$$\lim_{x \rightarrow -2^+} \frac{x^2-1}{x^2-4} = -\infty$$

$$\lim_{x \rightarrow -2^-} \frac{x^2-1}{x^2-4} = \infty$$

$$\frac{+}{-2^-} \frac{+}{-2^+}$$

Horizontal asymptotes

$$\lim_{x \rightarrow \pm\infty} \frac{x^2-1}{x^2-4} = 1$$

$y=1$ Horizontal asymptote

Roots

$x=1$

$x=-1$

$(1,0) (-1,0)$

$x=0$

$y=1/4$

$(0, 1/4)$

Symetry

$y(x) = y(-x)$

Even function symmetry y axis

$f'(x)$

$$y' = \frac{2x(x^2-4) - 2x(x^2-1)}{(x^2-4)^2} = \frac{2x^3 - 8x - 2x^3 + 2x}{(x^2-4)^2} = \frac{-6x}{(x^2-4)^2}$$

$y' = 0$

$x=0$ C.P.

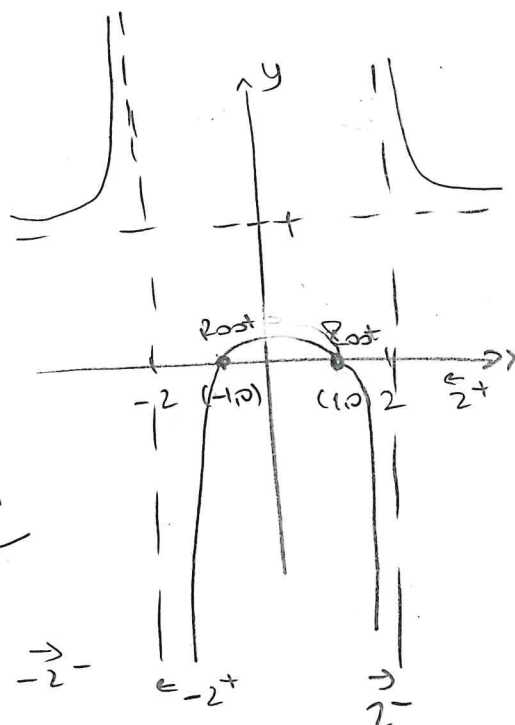
$x=\pm 2$ y' undef asymptotes

$f''(x)$

$$y'' = \frac{-6(x^2-4)^2 - 2(x^2-4) \cdot 2x(-6x)}{(x^2-4)^4} = \frac{6(3x^2+4)}{(x^2-4)^3}$$

$y'' \neq 0$

x	-2	0	2
y'	+	undef	-
y''	+	undef	+
y		max	



Ex Sketch the graph of $y = x \cdot e^{-x^2/2}$

$f(x)$ $D(f) = \mathbb{R}$

Asymptotes Horizontal $\lim_{x \rightarrow \pm\infty} \frac{x}{e^{x^2/2}} = \lim_{x \rightarrow \pm\infty} \frac{1}{e^{x^2/2} \cdot x} = 0$
 $y=0$ asymptote.

Vertical asymptote $\lim_{x \rightarrow \pm\infty} f(x) = \infty$ No vertical asymptote

Roots $y = \frac{x}{e^{x^2/2}}$ $x=0$ $y=0$ $(0,0)$

Symmetry $y(-x) = -x \cdot e^{-x^2/2} = -y(x)$ odd function
 Symmetry w.r.t. origin

$f'(x)$ $y' = e^{-x^2/2} - x \cdot e^{-x^2/2} \cdot x = (1-x^2)e^{-x^2/2}$

Critical points $x = \pm 1$ $y' = 0$

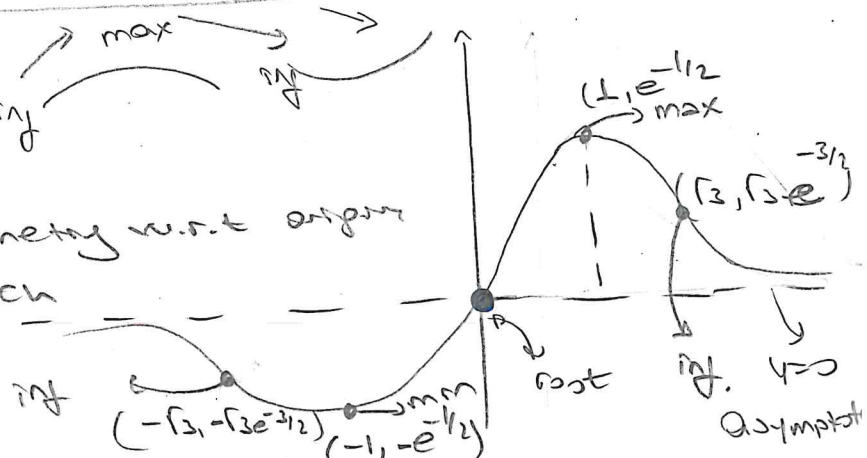
$f''(x)$ $y'' = -2x e^{-x^2/2} + (-x) \cdot e^{-x^2/2} (1-x^2)$
 $= e^{-x^2/2} (-2x - x + x^3) = e^{-x^2/2} (x^3 - 3x)$

$x(x^2 - 3) = 0$ $x=0$ $x = \pm\sqrt{3}$ $y''=0$ Inflection points

x	$-\sqrt{3}$	-1	0	1	$\sqrt{3}$
f'	—	0	+	0	—
f''	—	+	0	—	+

$f(1) = e^{-1/2}$
 $f(-1) = -e^{-1/2}$
 $f(\sqrt{3}) = \sqrt{3}e^{-3/2}$
 $f(-\sqrt{3}) = -\sqrt{3}e^{-3/2}$

(*) You can use symmetry w.r.t. origin after you sketch from one side.



Ex Sketch the graph of $f(x) = (x^2 - 1)^{2/3}$

$f(x)$ $D(f) = \mathbb{R}$

Asymptotes No vertical $(\lim_{x \rightarrow 0} f(x) = \infty)$
No horizontal $x \rightarrow \pm\infty \quad y \rightarrow \infty$

Roots $x = \pm 1, \quad y = 0 \quad (-1, 0) \quad (1, 0)$
 $x = 0 \quad y = 1 \quad (0, 1)$

Symmetry $f(-x) = ((-x)^2 - 1)^{2/3} = f(x)$ even symmetry wrt y.

$f'(x)$ $f'(x) = \frac{2}{3} (x^2 - 1)^{-1/3} \cdot 2x = \frac{4x}{3} (x^2 - 1)^{-1/3}$

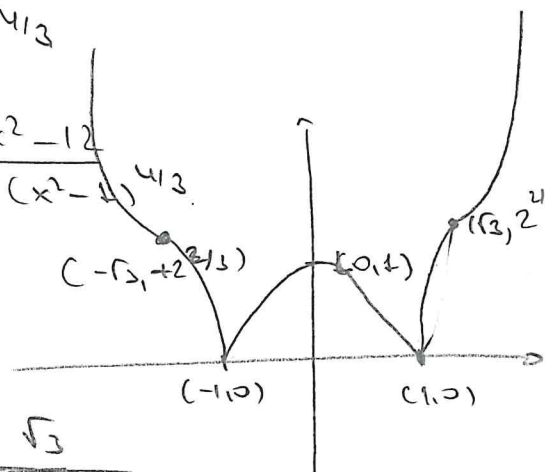
$x = 0 \Rightarrow f'(0) = 0$ C.P.

$x = \pm 1 \Rightarrow f'(\pm 1) = \text{undef}$ singular points.

$f''(x)$ $f''(x) = \frac{4}{3} (x^2 - 1)^{-1/3} + 2x (-1/3) (x^2 - 1)^{-4/3} \cdot \frac{4x}{3}$
 $= \frac{12}{9} (x^2 - 1)^{-1/3} - \frac{8x^2}{9} (x^2 - 1)^{-4/3}$

$= \frac{12(x^2 - 1) - 8x^2}{9(x^2 - 1)^{4/3}} = \frac{4x^2 - 12}{9(x^2 - 1)^{4/3}}$

$f''(x) = 0 \quad x = \pm \sqrt{3}$ Inf.
 $f''(x)$ undef at $x = \pm 1$



x	$-\sqrt{3}$	-1	0	1	$\sqrt{3}$
f'	—	undef	+	undef	+
f''	+	undef	—	undef	+
f	min	min	max	min	min

$f(-1) = 0 \quad f(1) = 0$

EXAMPLES

Ex. 1 Obtain extreme values of the following functions.

- (a) $f(x) = x^2$ $[-2, 1]$ (c) $f(x) = x^{2/3}$ $[-2, 3]$
 (b) $f(x) = 8x - x^4$ $[-2, 1]$ (d) $f(x) = \sqrt{4-x^2}$ $[-2, 1]$
 (e) $f(\theta) = \sin \theta$ $[-\pi/2, \pi/6]$ (f) $f(t) = 2 - 11t$ $[-1, 3]$

- (a) $f(x) = x^2$ $f'(x) = 2x$ ① Critical points (CP) $f'(x) = 0$ $x = 0$ $f(0) = 0$
 ② Singular points (SP) $f'(x)$ undef no points
 ③ End points (E.P) $f(-2) = 4$ $f(1) = 1$

We have abs. max at $x = -2$ $f(-2) = 4$
 abs min at $x = 0$ $f(0) = 0$.

(b) $f(x) = 8x - x^4$ $f'(x) = 8 - 4x^3$

- ① C.P $8 - 4x^3 = 0$ $x = \sqrt[3]{2} > 1$ not in $[-2, 1]$
 ② S.P No singular point
 ③ E.P $f(-2) = 0$ $f(1) = 7$
 abs min abs max

(c) $f(x) = x^{2/3}$ $f'(x) = \frac{2}{3 \cdot (x^{1/3})}$

- ① C.P $f'(x) = 0$ NO C.P
 ② S.P $x = 0$ $f(0) = 0$
 ③ E.P $f(-2) = \sqrt[3]{4}$ $f(3) = \sqrt[3]{9}$
 abs min abs max

(d) $f(x) = \sqrt{4-x^2}$ $f'(x) = \frac{-2x}{2\sqrt{4-x^2}}$

- ① C.P $x = 0$ $f(0) = \sqrt{4} = 2 \rightarrow$ abs max
 $x = 2 \notin [-2, 1]$
 ② S.P $x = \pm 2$ $f(\pm 2) = 0$ } abs min.
 ③ E.P $x = -2$ $f(-2) = 0$
 $x = 1$ $f(1) = \sqrt{3}$

② $f(\theta) = \sin \theta \quad [-\pi/2, 5\pi/6]$

① $f'(\theta) = \cos \theta = 0 \quad \theta = \pi/2$
 $f(\pi/2) = 1 \rightarrow \text{abs max}$

② $f'(\theta)$ undefined for no values

③ $f(-\pi/2) = \sin(-\pi/2) = -1 \rightarrow \text{abs min}$
 $f(5\pi/6) = \sin(5\pi/6) = \frac{1}{2}$

③ $f(t) = 2 - |t| \quad t \in [-1, 3]$
 $= 2 - \sqrt{t^2}$

① $f'(t) = -1/2 (t^2)^{-1/2} \cdot 2t = -\frac{t}{|t|}$
 C.P. $f'(t) = 0$ doesn't exist.

② $f'(t)$ undefined for $t = 0$
 $f(0) = 2 \rightarrow \text{abs max}$

③ E.P. $f(-1) = 1 \quad f(3) = -1$
 abs min

Ex 2 Obtain critical points, increasing, decreasing interval and local maximum values of the function.

① $f(x) = x^4 - 2x^2 - 3 \quad [-2, 2]$

② $f(x) = x - x^{2/3} \quad [-1, 2]$

③ $f(x) = x \sqrt{8 - x^2} \quad [-2\sqrt{2}, 2\sqrt{2}]$

④ $f(x) = \frac{x^2 - 3}{x - 2}$

⑤ $f(x) = x^{1/3} (x + 5)$

⑥ $f(x) = x^{2/3} (x + 5)$

⑦ $f(x) = \sqrt{4x^2 - x^3} \quad [-4, 4]$

① $f'(x) = 4x^3 - 4x$ $f'(x) = 0$ $x = 0$ $x = \pm 1$

	E.P	C.P	C.P		E.P
x	-2	-1	0	1	2
f'	-	0	+	0	-
f		local min	local max	local min	

local max $x = -2$

$f(0) = -3$
 $f(1) = -4$
 $f(-1) = -4$ } abs min
 $f(\pm 2) = 16 - 8 - 3 = 5$
 abs max

② $f'(x) = 1 - \frac{2}{3x^{1/3}} = \frac{3x^{1/3} - 2}{3x^{1/3}}$

	E.P	S.P	C.P	E.P
x	-1	0	8/27	2
f'	-	+	undef	+
f	local min	local max	local min	

$f(-1) = 4.667 > 0$
 $f'(0.12) = -0.375 < 0$

① $f'(x) = 0$ $3x^{1/3} = 2$
 $x = 8/27$ $f(8/27) = -\frac{4}{27}$

② $f'(x)$ undef at $x = 0$ $f(0) = 0$
 ③ E.P $f(-1) = -2 \rightarrow$ abs min
 $f(2) = 2 - 2 = 0.41$
 abs max

③ $f'(x) = \frac{x(-2x)}{2\sqrt{8-x^2}} = \frac{-2x^2}{\sqrt{8-x^2}}$

① $f'(x) = 0$ $x = \pm 2$ C.P $f(2) = 4$ $f(-2) = -4$
 abs max abs min
 ② $f'(x)$ undef $x = \pm 2\sqrt{2}$ S.P $f(\pm 2\sqrt{2}) = 0$
 ③ End points $f(\pm 2\sqrt{2}) = 0$

	E.P	C.P	C.P	E.P
x	$-2\sqrt{2}$	-2	2	$2\sqrt{2}$
f'	-	0	0	-
f		local min	local max	

④ $f(x) = \frac{x^2-3}{x-2}$ $f'(x) = \frac{2x(x-2) - (x^2-3)}{(x-2)^2} = \frac{2x^2 - 4x - x^2 + 3}{(x-2)^2}$

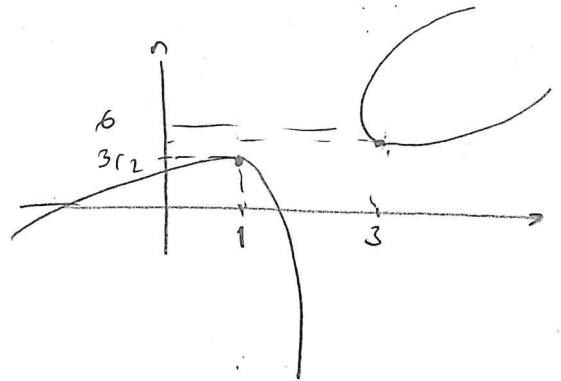
① C.P $f'(x) = 0$ $x = 3$ $x = 1$ $= \frac{x^2 - 4x + 3}{(x-2)^2} = \frac{(x-3)(x-1)}{(x-2)^2}$

② S.P $x = 2$

x	1	2	3
f'	+	0	-
		undef	0
			+

max → min

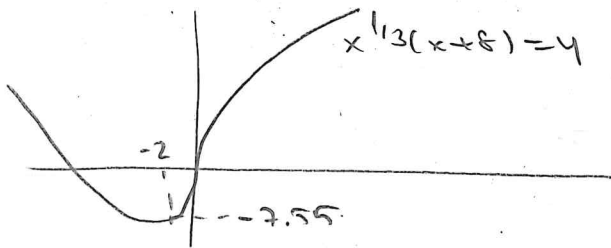
$\lim_{x \rightarrow \pm\infty} f(x) = \pm\infty$
 no abs max
 no abs min



⑤ $f(x) = x^{1/3}(x+8)$

$f'(x) = \frac{1}{3}x^{-2/3}(x+8) + x^{1/3}$

- ① $f'(x)=0 \quad x=-2 \quad f(-2) = \frac{6 \cdot (-2)^{1/3}}{3 \cdot (-2)^{2/3}} = \frac{4x+8}{3x^{2/3}}$
 ② $f'(x) = \text{undef} \quad x=0 \quad f(0)=0$
 ③ No end points



x	-2	0
f'	-	0
		+
		undef
		+

local min → local max

⑥ $f(x) = x^{2/3}(x+5) \quad f'(x) = \frac{2}{3}x^{-1/3}(x+5) + x^{2/3} = \frac{5x+10}{3x^{1/3}}$

① C.P $f'(x)=0 \quad x=-2$

② S.P $f'(x) \text{ undef} \quad x=0$

$\lim_{x \rightarrow \pm\infty} f(x) = \pm\infty$ No abs max
 abs min

*	-2	0
f'	+	0
		-
		undef
		+

local max → local min

⑦ $f(x) = \sqrt{4x^2 - x^3} \quad x \in [-4, 4]$

$f'(x) = \frac{8x - 3x^2}{2\sqrt{4x^2 - x^3}}$

① C.P $f'(x)=0 \quad x=0 \quad x=8/3$

② S.P $4x^2 - x^3 = 0 \quad x=0 \quad x=4$

x	E.P	0	8/3	E.P, S.P
	-4			4
f'	-		0	-
		undef	+	

abs min → abs max

$f(0)=0$
 $f(8/3) = 3.079$
 $f(4)=0$
 $f(-4) = 11.3137$
 abs max

Ex 3 Sketch the graph of following functions

1 $f(x) = \frac{x^2+1}{x+1}$

$f(x)$ $D_f = \mathbb{R} \setminus \{-1\}$ $f(-x) \neq -f(x)$ $f(-x) \neq f(x)$ } No symmetry

$x=0$ $y=1$ $(0,1)$

Asymptotes Horizontal $\lim_{x \rightarrow \pm\infty} \frac{x^2+1}{x+1} = \pm\infty$ No horizontal asymptote

Vertical $\lim_{x \rightarrow -1^+} \frac{x^2+1}{x+1} = \infty$
 $\lim_{x \rightarrow -1^-} \frac{x^2+1}{x+1} = -\infty$ } $x = -1$ vertical asymptote

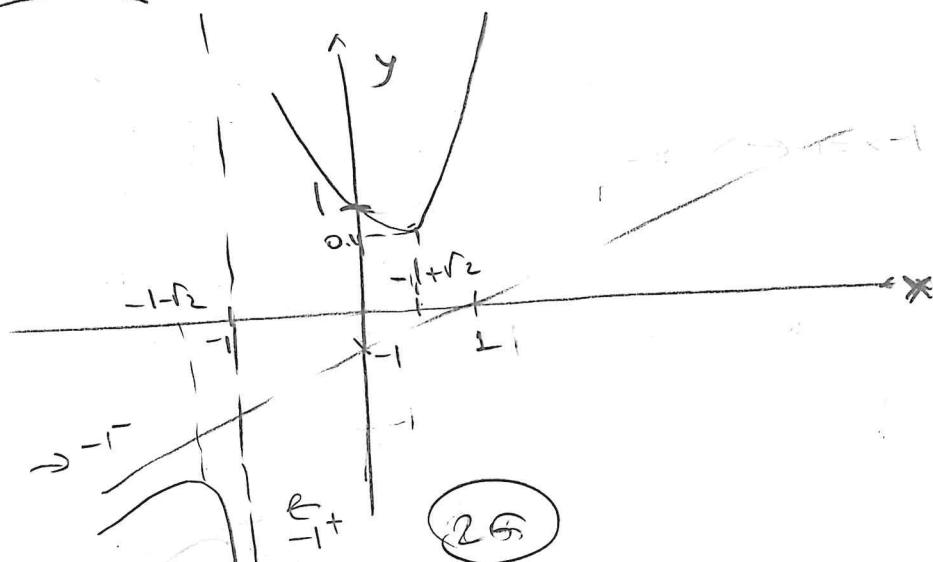
Oblique asymptote $\frac{x^2+1}{x+1} = \frac{x^2+x}{x+1} + \frac{-x+1}{x+1}$
 $y = x-1 + \frac{2}{x+1}$
 $y = x-1$ oblique asymptote

$f'(x)$ $f'(x) = \frac{x^2+2x-1}{(x+1)^2} = 0 \Leftrightarrow x_1 = -1+\sqrt{2}$
 $x_2 = -1-\sqrt{2}$
 ≈ 0.4
 ≈ -2.4

$f''(x) = \frac{4}{(x+1)^3} = y'' \neq 0$
 $x = -1$ sign changes

x	C.P $-1-\sqrt{2}$	S.P -1	C.P $-1+\sqrt{2}$
f'	+	undef	-
f''	-	undef	+

local max local min



(2) $f(x) = \frac{4}{x} - x$

$f(x)$ $D_f = \mathbb{R} \setminus \{0\}$

$f(-x) = -\frac{4}{x} + x = -f(x)$ Odd function
Symmetry w.r.t origin

$x=0$ $f(x)$ undef

$y=0$ $\frac{4-x^2}{x} = 0$ $x = \pm 2$ $(2,0)$ $(-2,0)$

$f'(x)$

$f'(x) = -\frac{4-x^2}{x^2}$

$f'(x) \neq 0$

$f'(x) < 0$ for all x . no root.

$x=0$ $f'(x)$ undef

$f''(x)$

$f''(x) = \frac{8}{x^3}$

$f''(x) = 0$

no root look if sign changes

changes write left of $x=0$

x	0		
f'	-	undef	-
f''	-	undef	+

Asymptotes

Horizontal

$\lim_{x \rightarrow \infty} \frac{4}{x} - x = \lim_{x \rightarrow \infty} \frac{4-x^2}{x} = \lim_{x \rightarrow \infty} -x = -\infty$

$\lim_{x \rightarrow -\infty} \frac{4}{x} - x = \infty$ No Horizontal

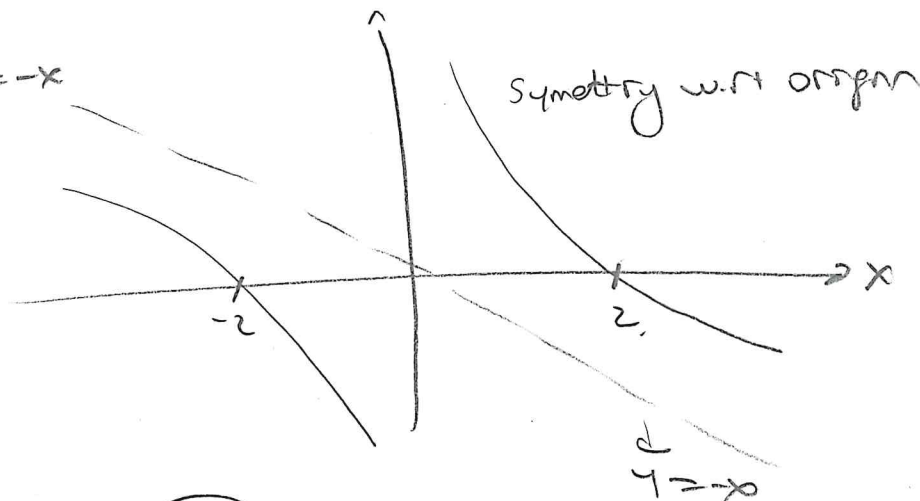
Vertical asymptote

$\lim_{x \rightarrow 0^-} \frac{4}{x} - x = \lim_{x \rightarrow 0^-} \frac{4-x^2}{x} = -\infty$

$\lim_{x \rightarrow 0^+} \frac{4}{x} - x = \lim_{x \rightarrow 0^+} \frac{4-x^2}{x} = \infty$

$x=0$
Vertical
Asymptote

Oblique asymptote $y = -x$



③ $f(x) = e^{1/x}$

$f(x)$ $D_f = \mathbb{R} \setminus \{0\}$

$f(x) \neq f(-x)$
 $f(x) \neq -f(-x)$ No symmetry

Asymptotes

Horizontal

$\lim_{x \rightarrow \pm\infty} e^{1/x} = 1$ $y=1$

Vertical

$\lim_{x \rightarrow 0^+} e^{1/x} = \infty$ $x=0$ vertical asym
 $\lim_{x \rightarrow 0^-} e^{1/x} = 0$

$f'(x)$

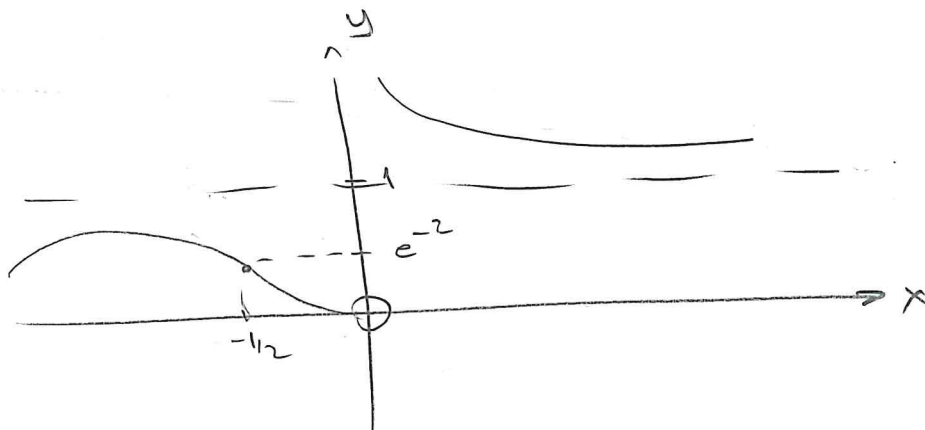
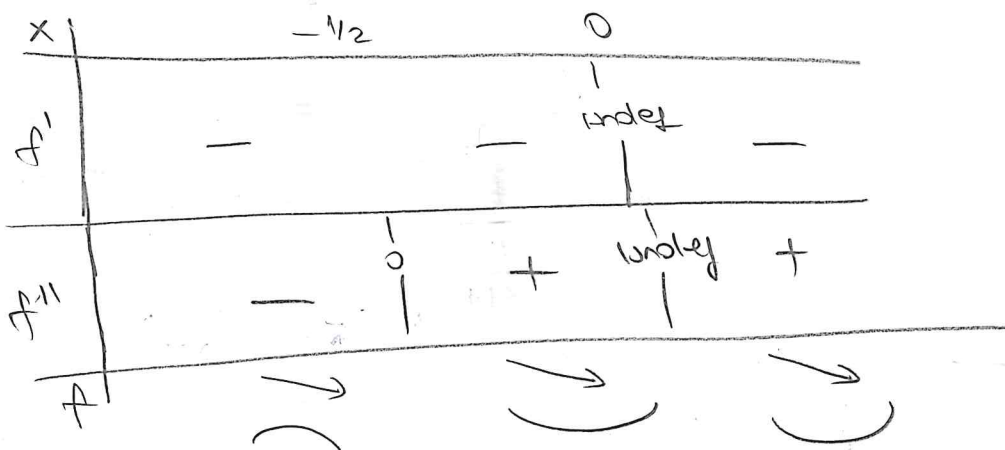
$f'(x) = e^{1/x} \left(-\frac{1}{x^2}\right) < 0$

S.P $x=0$ No. Critical points

$f''(x)$

$f''(x) = e^{1/x} \left(\frac{2x+1}{x^4}\right)$

$f''(-1/2) = 0$ $x=0$ undef



④ $f(x) = \frac{(x+1)^2}{x^2+1}$

$f(x)$ D_f = $\mathbb{R}$ $f(x) \neq f(-x)$ $-f(x) \neq f(-x)$ No symmetry
 $x=0$ $y=1$ $(0,1)$
 $x=-1$ $y=0$ $(-1,0)$

Asymptotes $f(x) = \frac{x^2+2x+1}{x^2+1} = 1 + \frac{2}{x} + \frac{1}{x^2}$
 $\frac{1}{x^2} \rightarrow 0$

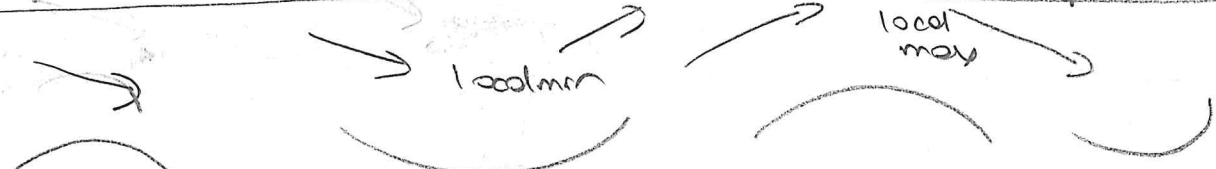
$\lim_{x \rightarrow \pm \infty} f(x) = 1$ $y=1$ HORIZONTAL ASYMPTOTE

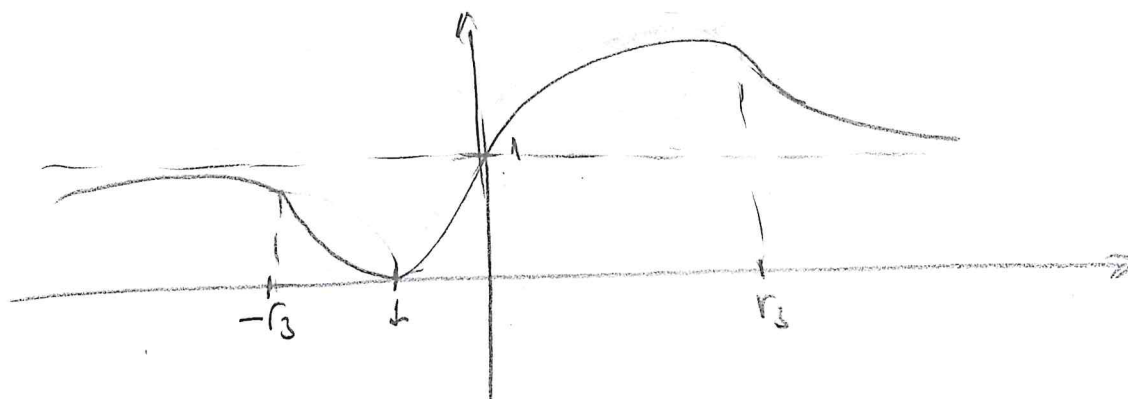
No vertical asymptote or oblique asymptote

$f'(x)$ $f'(x) = \frac{2(x+1)(1+x^2) - 2x(1+x)^2}{(1+x^2)^2} = \frac{2(1-x^2)}{(1+x^2)^2}$ C.P. $x = \pm 1$ No S.P.

$f''(x)$ $f''(x) = \frac{-4x(1+x^2)^2 - 2(1+x^2)2x \cdot 2(1-x^2)}{(1+x^2)^4} = \frac{4x(x^2-3)}{(1+x^2)^3}$
 $x=0$ $x = \pm \sqrt{3}$ inflection points

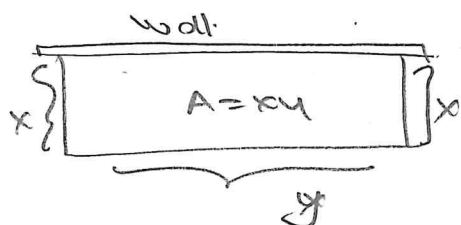
C.P. $x = \pm 1$ No singular points

x	$-\sqrt{3}$	-1	0	1	$\sqrt{3}$					
f'	-	-	0	+	+	+	0	-	-	
f''	-	0	+	0	-	-	0	+	+	
f										



EXTREME VALUE PROBLEMS

Ex A rectangular animal enclosure is to be constructed having one side along an existing long wall and the other three sides fenced. If 100m of fence are available, what is the largest possible area for the enclosure.



$$x + 2y = 100 \Rightarrow x = 100 - 2y$$

$$A(y) = (100 - 2y)y = 100y - 2y^2$$

Now $0 < y < 50$ (to area makes sense), the problem turns to finding the abs maximum of $A = 100y - 2y^2$ on $(0, 50)$.

$$A' = 100 - 4y = 0 \Rightarrow y = 25. \text{ The only critical point of } A \text{ is}$$

$$y = 25 \Rightarrow \text{largest area } A(25) = 1250 \text{ m}^2$$

Ex A lighthouse L is located on a small island 5 km north of a point A on a straight east-west shoreline. A cable is to be laid from L to point B on the shoreline 10 km east of A . The cable will be laid through the water in a straight line from L to a point C on the shoreline A and B , and from there to B along shoreline. The part of cable lying in the water cost \$5/km, and shoreline \$3/km.

Where C must be chosen to minimize the total cost

$$T = T(x) = 5\sqrt{25+x^2} + 3(10-x)$$

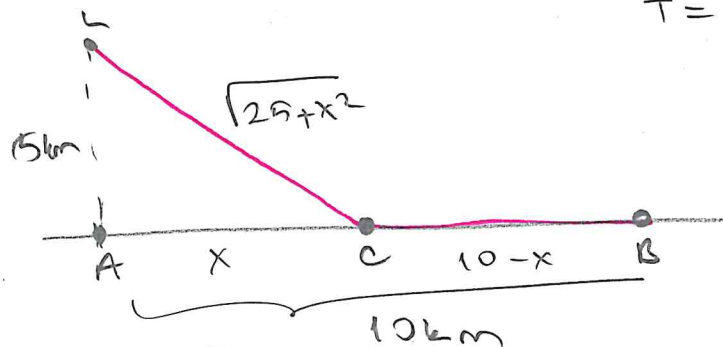
$$(0 \leq x \leq 10)_{EP}$$

$$T'(x) = \frac{5x}{\sqrt{25+x^2}} - 3 = 0$$

$$5x = 3\sqrt{25+x^2}$$

$$25x^2 = 9(25+x^2)$$

$$16x^2 = 225 \quad x^2 = \frac{15^2}{4} \quad x = \pm \frac{15}{4}$$



$$T(0) = 55$$

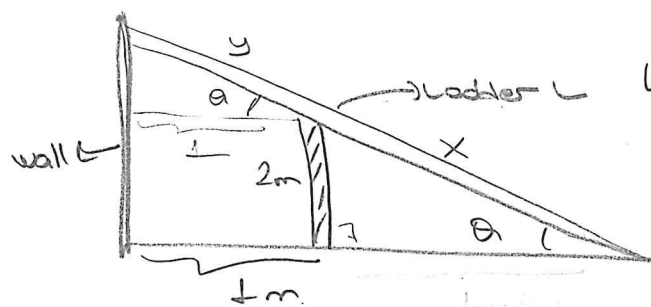
$$T(15/4) = 50$$

$$T(10) = 55.902$$

abs min

(29)

ex 7 Find the length of the shortest ladder that can extend from a vertical wall, over a fence 2m high located 1m away from the wall to a point on the ground outside the fence.



Length of the ladder: L

$$L = L(\theta) = \frac{1}{\cos \theta} + \frac{2}{\sin \theta}$$

$$0 < \theta < \pi/2$$

$$\frac{2}{x} = \sin \theta$$

$$x = \frac{2}{\sin \theta}$$

$$\frac{1}{y} = \cos \theta$$

$$y = \frac{1}{\cos \theta} \quad L = x + y$$

$$\lim_{\theta \rightarrow (\pi/2)^-} L(\theta) = \infty$$

$$\lim_{\theta \rightarrow 0^+} L(\theta) = \infty$$

$$\lim_{\theta \rightarrow 0^+} L(\theta) = \infty$$

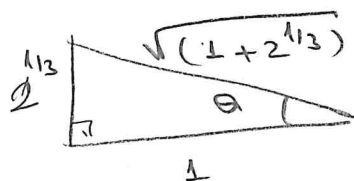
$$\theta \rightarrow 0^+$$

$L(\theta)$ must have a minimum value on $(0, \pi/2)$ occurs at a critical value

$$L'(\theta) = \frac{\sin \theta}{\cos^2 \theta} - \frac{2 \cos \theta}{\sin^2 \theta} = \frac{\sin^3 \theta - 2 \cos^3 \theta}{\cos^2 \theta \cdot \sin^2 \theta}$$

$$\sin^3 \theta = 2 \cos^3 \theta \Rightarrow \tan^3 \theta = 2$$

$$\tan \theta = 2^{1/3}$$



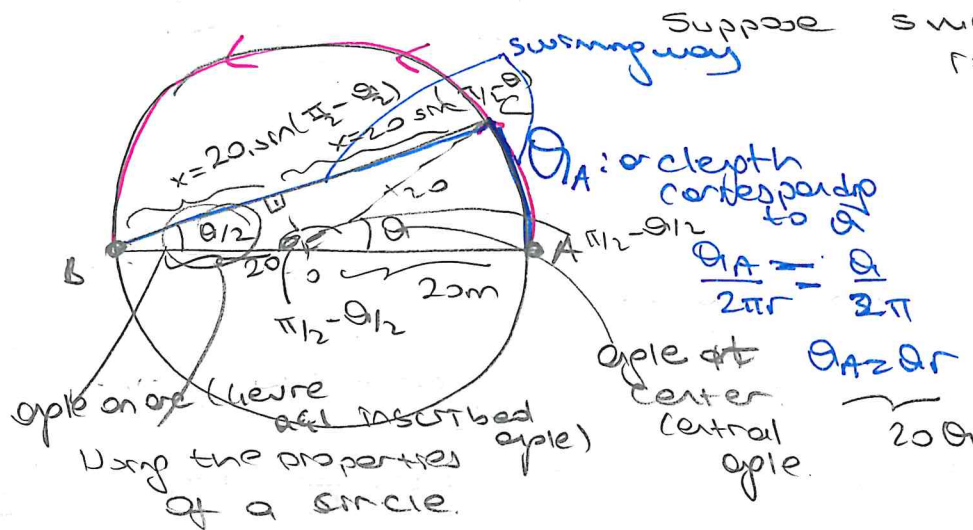
$$\cos \theta = \frac{1}{\sqrt{1 + 2^{1/3}}}$$

$$\sin \theta = \frac{2^{1/3}}{\sqrt{1 + 2^{1/3}}}$$

$$L(\theta) = \frac{1}{\cos \theta} + \frac{2}{\sin \theta} = (1 + 2^{1/3})^{1/2} + \frac{2(1 + 2^{1/3})^{1/2}}{2^{1/3}} \approx 4.16$$

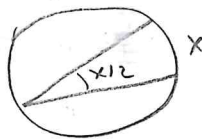
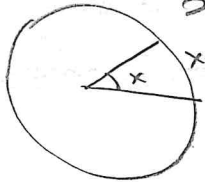
Ex A man can run twice as fast as he can swim. He is standing at a point A on the edge of circular swimming pool 40m in diameter, and he wishes to get diametrically opposite

point B as quickly as possible. He can run around the edge point C, then swim directly from C to B. Where should C be chosen to minimize the total time taken to get from A to B.



$$BC = 40 \cdot \sin\left(\frac{\pi - \theta}{2}\right)$$

RULES ON ANGLES OF CIRCLE



$$\frac{x}{2} = \sin\left(\frac{\pi - \theta}{2}\right)$$

$$t = t(\theta) = \text{time running} + \text{time swimming}$$

$$= \frac{20\theta}{2k} + \frac{40}{k} \cdot \sin\left(\frac{\pi - \theta}{2}\right)$$

$$0 \leq \theta \leq \pi$$

$$t'(\theta) = \frac{10}{k} - \frac{20}{k} \cos\left(\frac{\pi - \theta}{2}\right) = 0$$

$$\cos\left(\frac{\pi - \theta}{2}\right) = \frac{1}{2} \quad \frac{\pi - \theta}{2} = \frac{\pi}{3} \quad \theta = \frac{\pi}{3}$$

$$t\left(\frac{\pi}{3}\right) = \frac{10}{k} \left(\frac{\pi}{3} + \frac{4\sqrt{3}}{2} \right) \approx \frac{45.11}{k}$$

$$t(0) = \frac{40}{k}$$

$$t(\pi) = \frac{10\pi}{k} \approx \frac{31.4}{k}$$

(31)

6) LINEAR APPROXIMATIONS

Definition The linearization of the function f about a is the function L defined by

$$L(x) = f(a) + f'(a)(x-a) \approx f(x)$$

We say that $f(x) \approx L(x) = f(a) + f'(a)(x-a)$ provides linear approximation for values of f near a .

Ex 7 Find linearization of (a) $f(x) = \sqrt{1+x}$ about $x=0$ and

(b) $g(t) = 1/t$ about $t=1/2$

(a) We have $f(0) = 1$ since $f'(x) = \frac{1}{2\sqrt{1+x}}$ $f'(0) = 1/2$

The linearization of f about 0 is

$$L(x) = 1 + \frac{1}{2}(x-0) = 1 + \frac{x}{2}$$

(b) We have $g(1/2) = 2$ $g'(t) = -1/t^2$ $g'(1/2) = -4$. The linearization of $g(t)$ about $t=1/2$ is

$$L(t) = 2 - 4(t - 1/2) = 4 - 4t.$$

Ex 7 Use linearization to approximate the $\sqrt{26}$

Let $f(x) = \sqrt{x}$ then $f'(x) = \frac{1}{2\sqrt{x}}$ Since we know $f(25) = 5$

$f'(25) = 1/10$ then linearization of $f(x)$ about $x=25$ is

$$L(x) = f(25) + f'(25)(x-25)$$

$$= 5 + \frac{1}{10}(x-25)$$

$$L(26) = 5 + \frac{1}{10}(26-25) = 5 + \frac{1}{10} = 5.1$$

Exo 7 Approximate $\sqrt{4.2}$ using linear approximation of $\sqrt{1+x}$

$$f(x) = \sqrt{1+x}$$

$$f'(x) = \frac{1}{2\sqrt{1+x}}$$

$$f'(3) = \frac{1}{2\sqrt{1+3}} = \frac{1}{4}$$

To get $\sqrt{4.2}$ you must have $x = 3.2$

$$f(3) = \sqrt{1+3} = 2$$

$$L(x) = f(3) + f'(3)(x-3)$$

$$= 2 + \frac{1}{4}(x-3)$$

$$L(4.2) \approx f(3.2) = 2 + \frac{1}{4}(3.2-3) = 2 + \frac{0.2}{4} = 2.050$$

$$\text{Exact value of } \sqrt{4.2} = 2.04939$$

NOTE This question can be ask just $\sqrt{4.2} \approx ?$

That time function depends on the student
for example ① $f(x) = \sqrt{x}$ can be choosen with $x_0 = 4$.

$$f(4) = \sqrt{4} = 2$$

$$f'(4) = \frac{1}{2\sqrt{4}} = \frac{1}{4}$$

$$L(x) = f(4) + f'(4) \cdot (x-4)$$

$$L(4.2) = 2 + \frac{1}{4}(4.2-4) = 2 + 0.05$$

② $f(x) = \sqrt{1+x}$ $x=0$ can be choosen.

$$L(x) = f(0) + f'(0) \cdot (x-0)$$

$$= 1 + \frac{1}{2\sqrt{1+0}}(x-0) = 1 + \frac{1}{2}x$$

$$L(4.2) = 1 + \frac{1}{2} \cdot (3.2) = 1 + 1.6 = 2.6$$

What is the difference if you choose a near value to 4.2 you obtain more appropriate solution.